

Let's start by adding the tracepoint call in our target function `handle_newconnect()`. I have created my own repos of clean and instrumented versions with minor edits (which allow it to get compiled properly), but original versions can be found on <http://www.acme.com> (Jef Poskanzer).

```
wget http://step.polymtl.ca/~suchakra/osfy/thttpd-clean.tar.gz
tar -xvf thttpd-clean.tar.gz && cd thttpd-clean
```

For instrumenting the function, there are three basic steps. These are given below.

## Tracepoint declaration

First, prepare the *tracepoint provider* header. This provides a tracepoint declaration in which you can specify the tracepoint provider (TP) name, arguments, payload variable types, fields and expressions. Refer to the *ltng-ust man* page for detailed field types that you can use. Some examples are *ctf\_integer*, *ctf\_float* and *ctf\_string*. The `TRACEPOINT_EVENT` macro sets everything up for you once you specify the arguments. We can create this header for our tracepoint and name it as `thttpd_tp_provider.h`

```
/*
 * thttpd_tp_provider.h
 */

#undef TRACEPOINT_PROVIDER
#define TRACEPOINT_PROVIDER thttpd

#undef TRACEPOINT_INCLUDE_FILE
#define TRACEPOINT_INCLUDE_FILE ../thttpd_tp_provider.h

#ifdef __cplusplus
extern "C"{
#endif /* __cplusplus */

#if !defined(_THHTTPD_PROVIDER_H) || defined(TRACEPOINT_HEADER_MULTI_READ)
#define _THHTTPD_PROVIDER_H
#include <ltng/tracepoint.h>
TRACEPOINT_EVENT(
    thttpd,                                /* tracepoint provider
name */
    count,                                /* tracepoint name */
    TP_ARGS(int, var),
    TP_FIELDS( ctf_integer(int, count, var))
)
#endif /* _THHTTPD_PROVIDER_H */

#include <ltng/tracepoint-event.h>

#ifdef __cplusplus
}
```

```
#endif /* __cplusplus */
```

We also need to create a C file in which `TRACEPOINT_CREATE_PROBES` is defined. This will get the trace probes ready.

```
/*
 * thttpd_tp.c
 */
#define TRACEPOINT_CREATE_PROBES
#include "thttpd_tp_provider.h"
```

## Tracepoint call

The next step is to add the tracepoint to the target function. Insert the following lines in `thttpd.c` above the `main()`:

```
#define TRACEPOINT_DEFINE
#include "thttpd_tp_provider.h"
```

Insert the tracepoint functions inside the `handle_newconnect()` body. I choose to put it in such a way as to keep track of the `http_conn_count` while I trace.

```
handle_newconnect(struct timeval *tvP, int listen_fd)
{
    .
    .
    .

}
c->hc->initialized = 0;
++httpd_conn_count;

/* tracepoint with a payload as the connection count
*/
    tracepoint(thttpd, count, httpd_conn_count);
}
/* Get the connection. */
switch (httpd_get_conn(hs, listen_fd, c->hc)) {
    .
    .
    .
}
```

## Building it all

Once this is done, add the `thttpd_tp.c` file to the SRC variable in `Makefile.in`, and add `ltng-ust` library and `libdl` to the linking process while building the source.

```
export LDFLAGS="-ldl -ltng-ust"
./configure
make
```

Now you have completed the instrumentation phase of the application. Whenever this instrumented `thttpd` starts, it will